

4. Project Planning Phase

Introduction

Project planning is an essential phase that defines the roadmap for successful project completion. It includes task allocation, timeline estimation, milestone planning, and resource management. Proper planning ensures that every team member understands their responsibilities and that the project progresses in a systematic manner.

Project Objectives

- Develop a Machine Learning model to recommend the most suitable crop.
- Build a Flask-based web application with full user authentication.
- Perform data preprocessing and visualization.
- Evaluate multiple models using classification metrics.
- Deploy a user-friendly, secure crop recommendation system.

Team Members and Responsibilities

Team Member	Responsibility
Pinjari Arshiya (Team Lead)	Project Workflow, Environment Setup, Flask Integration, Auth Blueprint, GitHub Management
Muthukuri Raghu	Dataset Collection, Data Exploration, Data Visualization
Saisriram Tiruveedhi	Entity-Relationship Diagram, Data Preprocessing, Feature Selection
Shiva Shankar Vara Prasad Kuruva	Model Training, Model Evaluation, Model Saving
Uday Kiran	Flask Backend, Testing

Sprint Planning

Sprint 1

- Install required libraries
- Create project structure
- Download crop recommendation dataset

Sprint 2

- Data preprocessing
- Data visualization
- Feature selection

Sprint 3

- Train and compare Logistic Regression, Random Forest, KNN, SVC and Naive Bayes models
- Evaluate performance
- Save best-performing model

Sprint 4

- Develop Flask application
- Design HTML pages

- Perform testing
- Upload to GitHub

Project Timeline

Week 1

- Requirement Analysis
- Brainstorming

Week 2

- Dataset Collection
- Data Exploration

Week 3

- Model Development

Week 4

- Flask Deployment
- Documentation

Week 5

- Testing

Milestones

- ✓ Environment Setup
- ✓ Dataset Collection
- ✓ Data Preprocessing
- ✓ Model Training
- ✓ Model Evaluation
- ✓ Model Saving
- ✓ Flask Deployment
- ✓ Documentation
- ✓ Final Demonstration

Conclusion

The planning phase ensured efficient task distribution, proper time management, and successful completion of the project within the specified timeline.